DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Mini Project Report 22AIL45
on

COVID-19 Report Analysis

Submitted by

BhagyaShree Shegunasi	**1DS25AI400**
Nagesh K	**1DS25AI401**
Sahana S	**1DS25AI402**
Vardhan Reddy N S	**1DS25AI405**

Department of Artificial Intelligence & Machine Learning
DSCE, Bengaluru

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANASANGAMA, BELAGAVI-590018, KARNATAKA, INDIA
2025-26

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

CERTIFICATE

This is to certify that the Mini Project work titled "**COVID-19 Report Analysis**", carried out as part of the AAT Component of the **Data Visualization Laboratory**, has been successfully completed and submitted by BhagyaShree Shegunasi (1DS25AI400), Nagesh K (1DS25AI401), Sahana S (1DS25AI402), Vardhan Reddy N S (1DS25AI405). This report is a bonafide record of the Mini Project undertaken by the above students as a part of their academic curriculum under our supervision. It is submitted towards the partial fulfilment of the requirements for the award of the Bachelor of Engineering Degree in Artificial Intelligence and Machine Learning during the Fourth Semester of the academic year 2025–2026.

It is further certified that all the suggestions, modifications, and corrections recommended during the internal assessment have been duly incorporated into the report. The Mini Project work has been evaluated and approved, as it satisfies the academic requirements and conforms to the rules and regulations prescribed for the Bachelor of Engineering Degree program.

Course Coordinator	**Head of Department**
Ensteih Silvia	**Dr. Vindhya P Malagi**
Assistant Professor	**(HOD, Dept. of AI & ML)**
DSCE, Bengaluru	**DSCE,Bengaluru**

Deepshree Buchade
Assistant Professor
DSCE, Bengaluru

Table of Contents

INTRODUCTION

The COVID-19 pandemic emerged as a major global health crisis that affected millions of people worldwide. Since its outbreak in late 2019, the virus created significant challenges for healthcare systems, economies, and societies. To understand and manage the spread of the disease, governments and health organizations relied on data analysis and visualization to track cases, deaths, testing, and vaccination progress.

This project, **"COVID-19 Data Visualization Using Power BI,"** aims to analyze COVID-19 data and present meaningful insights through interactive dashboards. The project uses the **Our World in Data (OWID) COVID-19 dataset**, which contains comprehensive information collected from official sources. Microsoft Power BI was selected because of its ability to transform large datasets into clear and interactive visual reports.

The dashboard includes various visualizations such as KPI cards, line charts, bar charts, maps, and pie charts to display important pandemic trends. Interactive filters allow users to explore data based on different time periods and regions. The project helps users understand the impact of COVID-19, identify key trends, and demonstrate the role of business intelligence tools in public health analysis.

PROBLEM STATEMENT

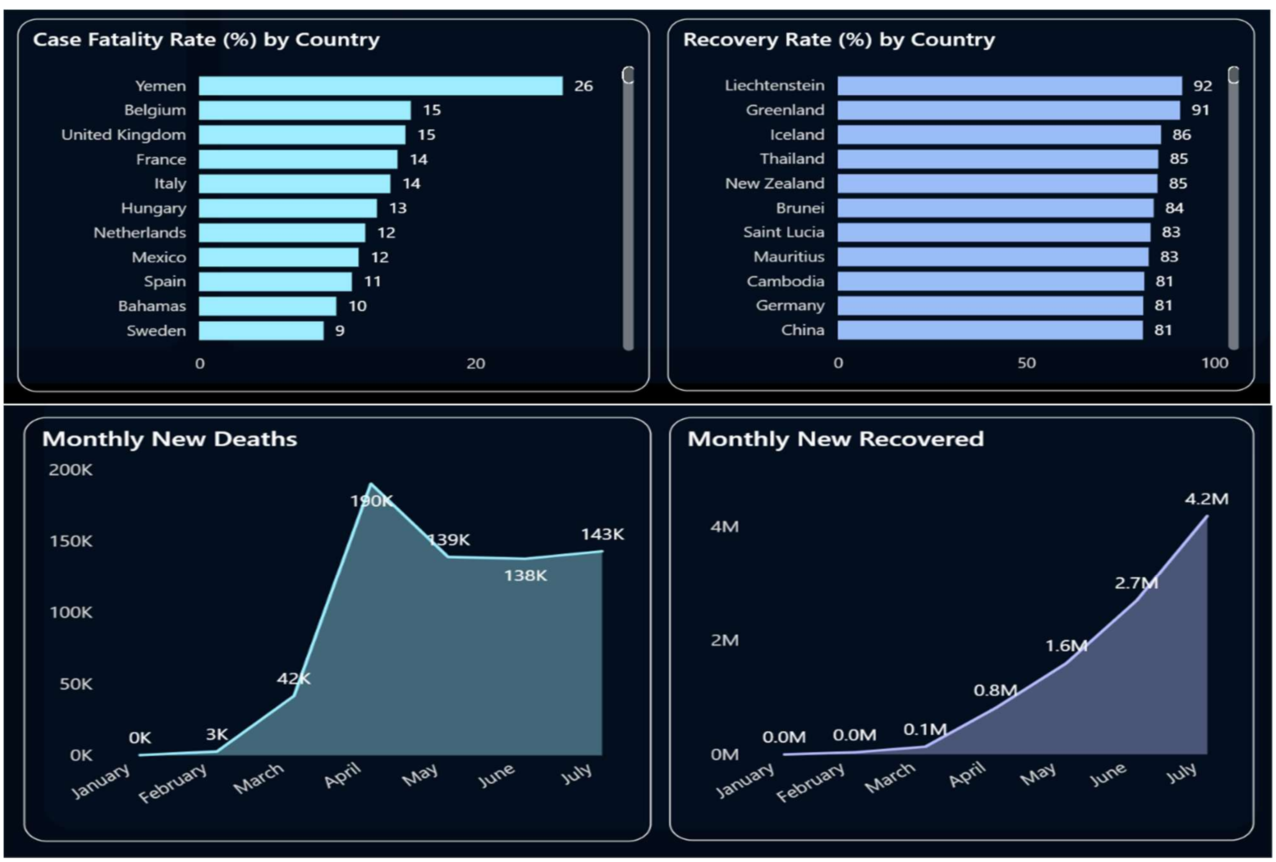
The COVID-19 pandemic produced a vast amount of data related to confirmed cases, deaths, testing, and vaccinations. Although this information is publicly available, it is often stored in large datasets that are difficult to understand in raw tabular form. As a result, identifying trends, patterns, and key insights becomes challenging for decision-makers, researchers, and the general public.

India experienced multiple waves of COVID-19 with varying levels of impact over time. Monitoring important indicators such as infection rates, mortality rates, testing activities, and vaccination coverage is essential for understanding the progression of the pandemic. However, without proper visualization tools, analyzing these factors and identifying critical events becomes difficult.

This project addresses the need for a clear and interactive platform to visualize COVID-19 data. Using Microsoft Power BI, the dashboard transforms raw data into meaningful visual insights through charts, maps, KPIs, and filters. It enables users to explore pandemic trends, analyze vaccination progress, and understand the relationship between vaccination rates and mortality, supporting better awareness and data-driven decision-making.

SYSTEM ARCHITECHTURE & DESIGN


COVID-19 Country Breakdown
Overview
Trends
Countries
Data Table
Total Confirmed Cases
829M
Total Deaths
43M
Total Recovered
388M
Total Active Cases
397M
Fatality Rate(CFR)
5.24
Active Cases by Country
Confirmed vs Recovered by Country
Confirmed - Deaths - Recovered (Top 15)
Country Vise Active
Country/Region, ...
Afghanistan
Albania
Algeria
Andorra
Angola
Antigua and Barb...
Argentina
Armenia
Australia
Austria
Deaths vs Recovered by Country
Case Fatality Rate (%) by Country
Recovery Rate (%) by Country


Case Fatality Rate (%) by Country
Yemen 26
Belgium 15
United Kingdom 15
France 14
Italy 14
Hungary 13
Netherlands 12
Mexico 12
Spain 11
Bahamas 10
Sweden 9
0
20
Recovery Rate (%) by Country
Liechtenstein 92
Greenland 91
Iceland 86
Thailand 85
New Zealand 85
Brunei 84
Saint Lucia 83
Mauritius 83
Cambodia 81
Germany 81
China 81
0
50
100
Monthly New Deaths
200K
150K
100K
50K
0K
0K
3K
42K
190K
139K
138K
143K
January
February
March
April
May
June
July
Monthly New Recovered
4M
2M
0M
0.0M
0.0M
0.1M
0.8M
1.6M
2.7M
4.2M
January
February
March
April
May
June
July

SDG MAPPING

- **SDG 3 – Good Health and Well-Being**
  - Sub-SDG 3.3 – Combat communicable diseases by monitoring and analyzing COVID-19 cases, deaths, and infection trends.
  - Sub-SDG 3.8 – Support access to quality healthcare information through effective visualization of public health data.

- **SDG 9 – Industry, Innovation and Infrastructure**
  - Sub-SDG 9.5 – Promote innovation and technological solutions by utilizing business intelligence tools for healthcare data analysis and reporting.

- **SDG 17 – Partnerships for the Goals**
  - Sub-SDG 17.18 – Improve the availability and accessibility of high-quality data through COVID-19 data visualization and analytics for informed decision-making.

- **SDG 11 – Sustainable Cities and Communities**
  - **Sub-SDG 11.B** – Support effective disaster and public health risk management by providing insights pandemic trends and vaccination progress.

APPLICATION

• Pandemic Trend Analysis: Helps identify changes in COVID-19 cases, deaths, testing, and vaccination rates over time and understand the progression of the pandemic.

• Public Health Monitoring: Assists healthcare organizations and government agencies in tracking key health indicators and monitoring the impact of COVID-19.

• Vaccination Progress Tracking: Enables analysis of vaccination coverage and helps assess the effectiveness of vaccination campaigns.

• Policy Making and Planning: Supports policymakers in developing data-driven strategies for disease prevention, healthcare management, and emergency response.

• Public Awareness: Provides clear and interactive visualizations that help citizens understand pandemic trends and health-related information.

• Data-Driven Decision Making: Transforms complex COVID-19 datasets into meaningful insights, enabling informed decisions by researchers, healthcare professionals, and authorities.

• Healthcare Resource Management: Helps identify periods of high infection rates, allowing better allocation of medical resources, hospital facilities, and healthcare services.

IMPLEMENTATION

The implementation of the **COVID-19 Data Visualization Dashboard** was carried out using Microsoft Power BI. The COVID-19 dataset was imported from the **Our World in Data (OWID)** source and contained information related to confirmed cases, deaths, testing, vaccinations, population, and other pandemic-related indicators. The dataset was processed and prepared for analysis to generate meaningful insights.

In the initial stage, data preprocessing was performed using Power Query. Missing values, duplicate records, and inconsistencies were identified and handled to improve data quality. After cleaning and transforming the data, it was loaded into Power BI for analysis. Various DAX measures were created to calculate key metrics such as total cases, total deaths, total vaccinations, death rate, and vaccination rate.

Multiple interactive dashboards were developed using KPI cards, line charts, bar charts, maps, pie charts, and scatter plots. Slicers were added to allow users to filter data by year, month, day, and continent for dynamic exploration. The final dashboard provides a comprehensive view of COVID-19 trends, vaccination progress, and mortality patterns, demonstrating how Power BI can transform large healthcare datasets into valuable insights for analysis and decision-making.

REFERENCES

1. Our World in Data (OWID) COVID-19 Dataset Our World in Data
   https://ourworldindata.org/coronavirus
2. COVID-19 Data Analysis and Visualization Using Power BI
   Microsoft Power BI Documentation
   https://learn.microsoft.com/en-us/power-bi/
3. World Health Organization (WHO) – Coronavirus (COVID-19) Dashboard
   World Health Organization
   https://covid19.who.int/
4. Ministry of Health and Family Welfare (MoHFW), Government of India
   https://www.mohfw.gov.in/
5. United Nations Sustainable Development Goals (SDGs)
   https://sdgs.un.org/goals
6. United Nations Sustainable Development Goal 3 – Good Health and Well-Being
   https://sdgs.un.org/goals/goal3
7. DAX Functions Reference – Microsoft Learn
   https://learn.microsoft.com/en-us/dax/
8. Power Query Documentation – Microsoft Learn
   https://learn.microsoft.com/en-us/power-query/

CO-PO PSO Mapping

CO1	Apply data cleaning, transformation, and modelling techniques in Power BI to prepare datasets for analysis
CO2	Analyse datasets using Power BI visualizations to identify trends, patterns, and relationships
CO3	Evaluate the effectiveness of different visualization methods for representing data accurately and clearly.
CO4	Design comprehensive, interactive dashboards in Power BI to effectively present analytical results.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	POS2	PSO3
1	3				3								3	
2		3			3								3	
3		3			3								3	
4			3	2	3								3	